AP Chemistry: Atomic Theory III **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) What is the electron configuration for nitrogen in its ground state? (long form)

1

2) What is the electron configuration for manganese in its ground state? (long form)

2

3) What is the ground state electron configuration of the iodine ion, I^-? (long form)

3

4) What is the ground state electron configuration of palladium? (short form)

4

5) What is the ground state electron configuration of lead? (short form)

5

6) What is the ground state electron configuration of rutherfordium? (short form)

6

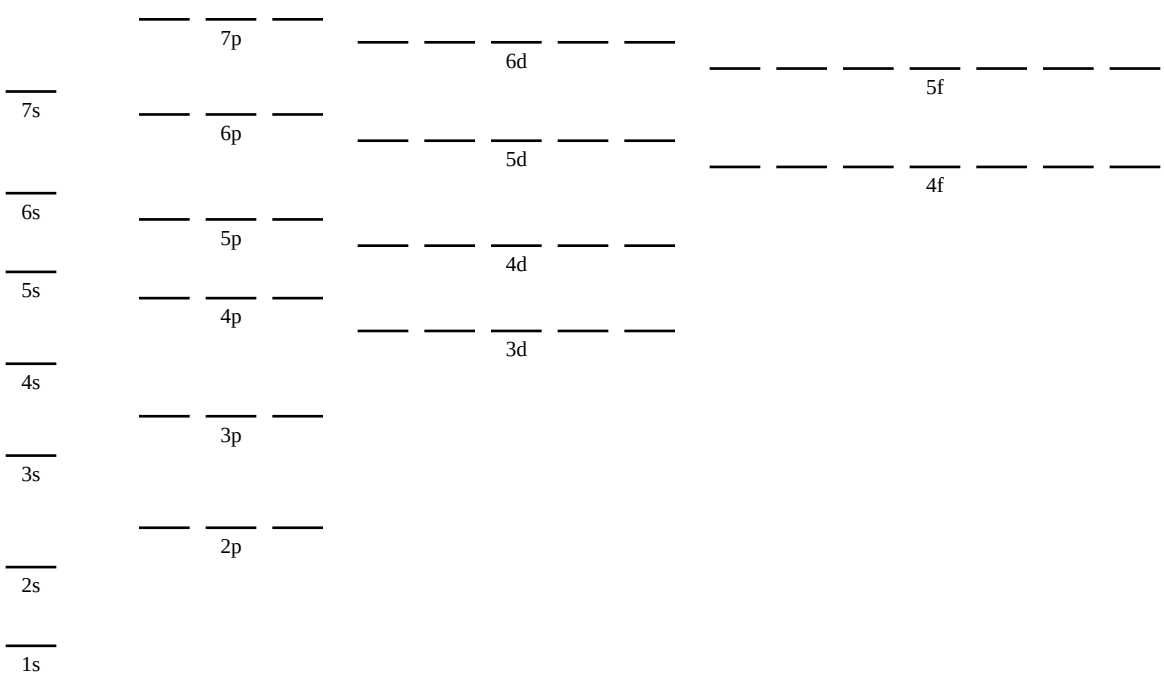
7) What is the ground state electron configuration of plutonium? (short form)

7

8) The following questions pertain to the sulfur element.

a. Draw an orbital diagram for sulfur in its ground state.

8a

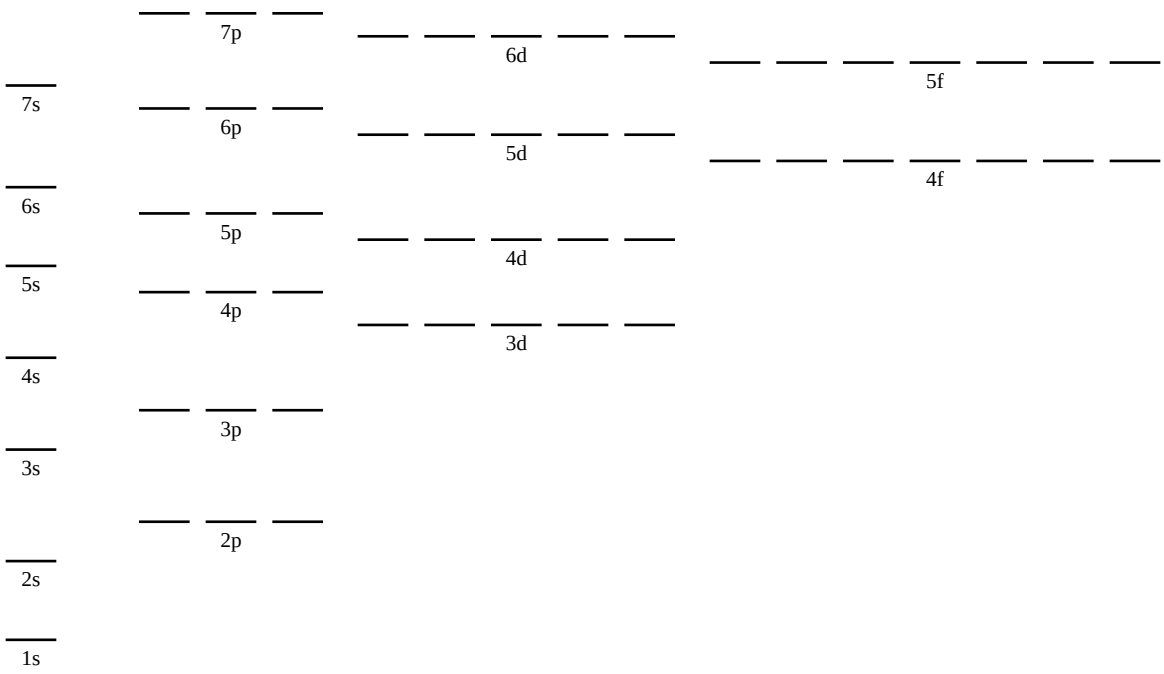


b. Does sulfur have a magnetic moment? Justify your answer.

8b

9) Draw an orbital diagram for iron in its ground state.

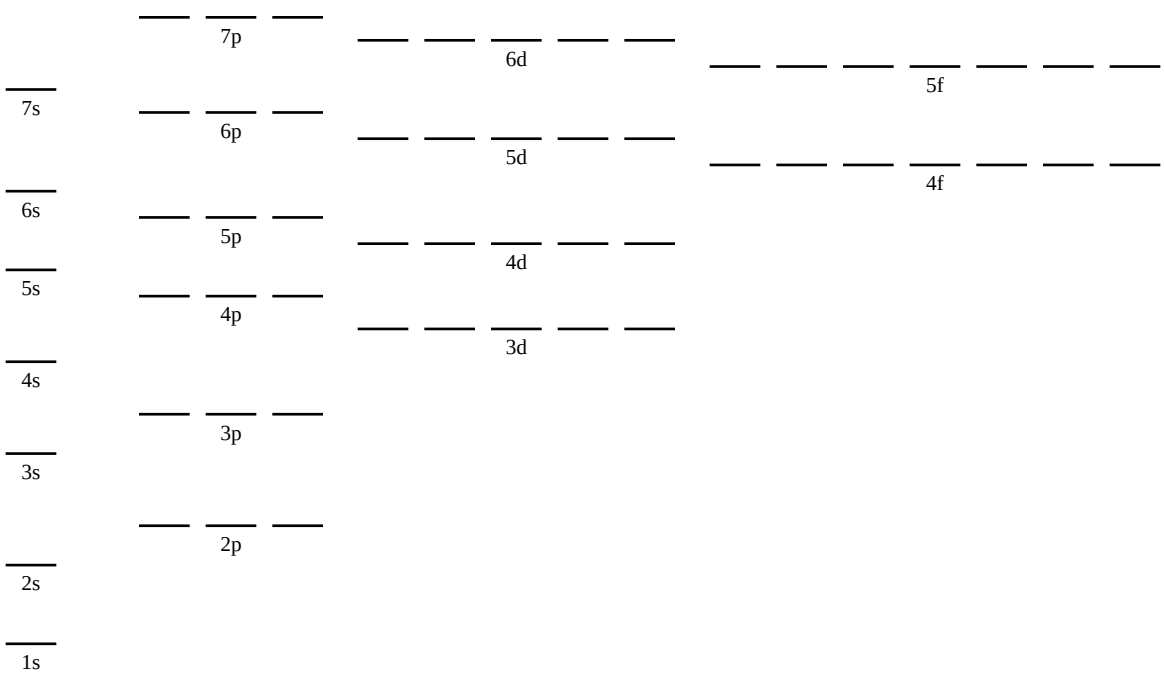
9



10) The following questions pertain to the element calcium.

a. Draw the orbital diagram for calcium in its ground state.

10a



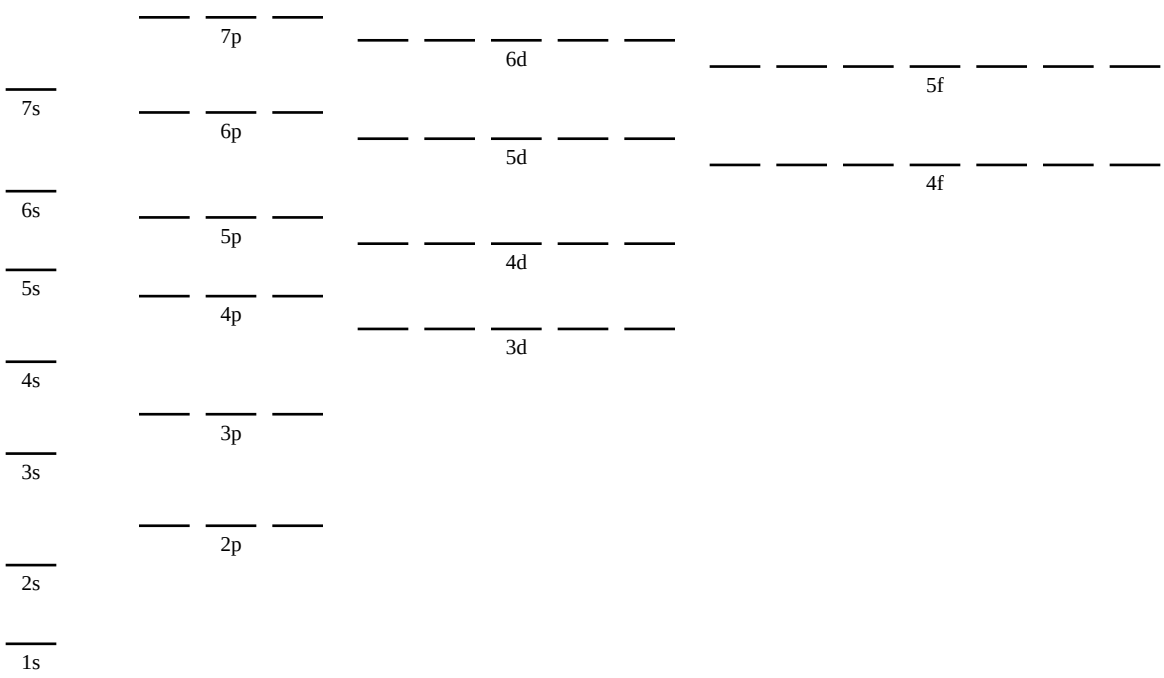
b. Is calcium paramagnetic or diamagnetic? Justify your answer.

10b

11) The following questions pertain to the silicon element.

a. Draw the orbital diagram for silicon in its ground state.

11a



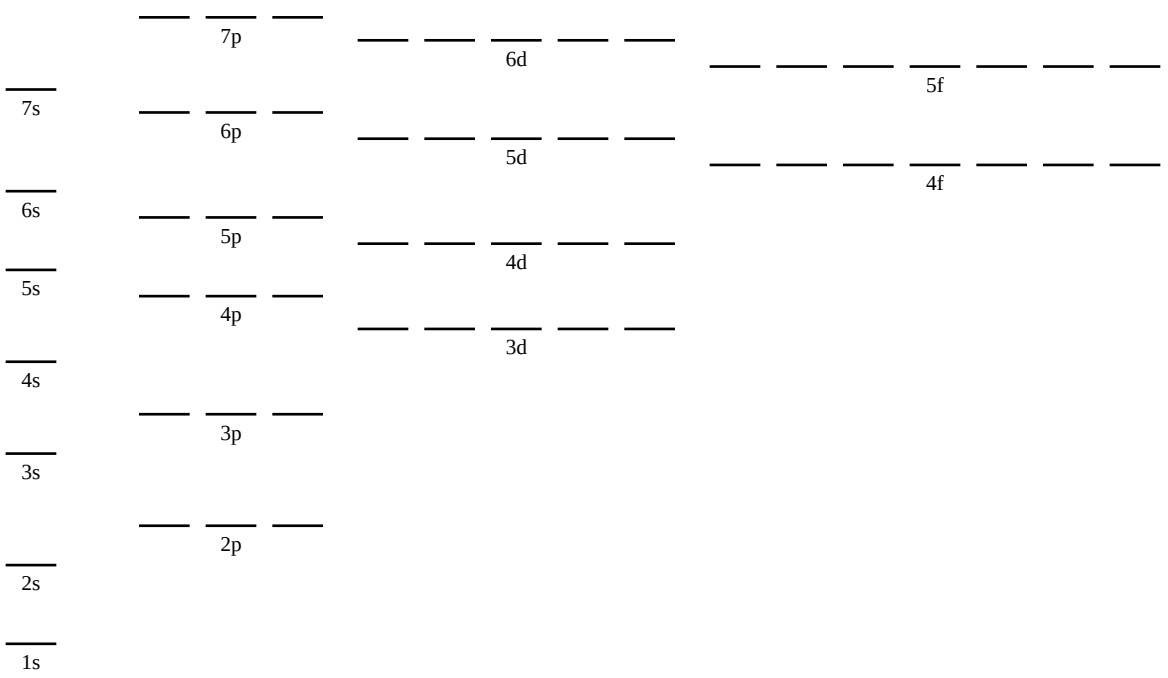
b. Is silicon paramagnetic or diamagnetic? Justify your answer.

11b

12) The following questions pertain to the carbon element.

a. Draw an orbital diagram for carbon in its ground state.

12a



b. Is carbon paramagnetic or diamagnetic? Justify your answer.

12b

13) Which atom in each set has the larger radius?

a. Be or O

13a

b. Cu or Br

13b

c. F or I

13c

d. O or As

13d

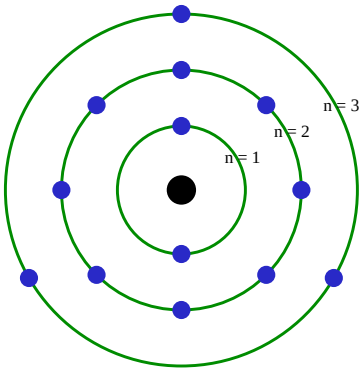
e. Kr or K

13e

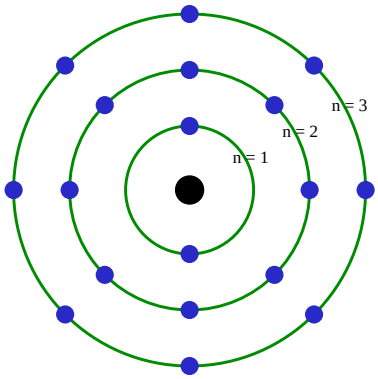
f. Ba or Li

13f

14) The shell models of two neutral atoms are drawn below. The representations are not drawn to scale. Which atom has the smallest radius? Justify your answer.



Element A



Element B

14

15) Explain each of the following occurrences by referring to the structure of the atoms in question (energy levels, orbitals, protons, etc.).

a. The atomic radius of oxygen is smaller than the atomic radius of carbon.

15a

b. The atomic radius of Mg is smaller than the atomic radius of Ca.

15b

16) Which of the following are isoelectronic:

a. Ne and F^-

16a

b. Ca^{2+} and Se^{2-}

16b

c. N and F^-

16c

d. I^- and Ba^{2+}

16d

e. K^+ and Ca^{2+}

16e

17) Mg^{2+} and F^- are isoelectronic.

a. Which ion has the smaller radius?

17a

b. Explain why the radii of these two ions are different sizes. Justify your claims.

17b

18) Which species has a smaller radius: Ca or Ca^{2+}? Provide an explanation and justify your claims using your knowledge of atomic structure.

18

19) Which species has a smaller radius: F or F^-? Provide an explanation and justify your claims using your knowledge of atomic structure.

19

20) Explain each of the following occurrences by referencing the structure of the atoms in question (energy levels, orbitals, protons, etc.).

a. The first ionization energy of Mg is 738.1 kJ/mol, while the first ionization energy of Al is only 577.9 kJ/mol.

20a

b. The first ionization energy of Mg is greater than the first ionization energy of Ca.

20b

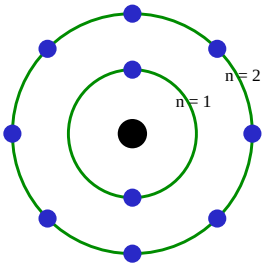
c. The first ionization energy of Na is less than the first ionization energy of Cl.

20c

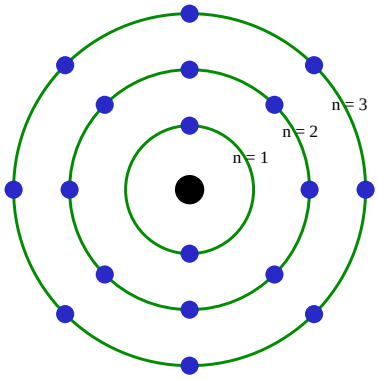
d. The second ionization energy of Na is 4560 kJ/mol, while the second ionization energy of Mg is only 1450 kJ/mol.

20d

21) The shell models of two neutral atoms are drawn below. The representations are not drawn to scale. Which atom would have the highest first ionization energy? Justify your answer.



Element A



Element B

21

22) Which element from each set is most electronegative?

a. F or C

22a

b. Al or Cl

22b

c. Po or S

22c

d. Cs or I

22d

e. Ca or Cl

22e

f. O or Se

22f

g. Zn or K

22g

h. C or Pb

22h

i. Ga or O

22i

23) Sulfur is more electronegative than calcium. Explain why this is and justify your claims using your knowledge of atomic structure.

23